

Technical Note

DIRECT ESTERIFICATION

DESCRIPTION

Dorf Ketal Tyzor® organic titanates are excellent catalysts in a number of chemical reactions. They are particularly useful in esterifications and transesterifications because the catalysis is accomplished with a minimum of undesirable side reactions and results in a high conversion rate at a high yield of esters with very low color and odor formation. For example, esters can be made from carboxylic acids and secondary alcohols with only a minor amount of olefin formation, in contrast to conventional acid catalysis, which produces substantial amounts of olefins.

Titanates are particularly useful for reactions involving unsaturated reactants and amino alcohols. Similarly, titanates are condensation catalysts in the manufacture of polyesters. Organic titanates are useful for catalysis and cross-linking of many other resin types, including polyolefins, polyurethanes, epoxies, phenolics and silicones. Organic titanates also may be used as catalysts for aldol condensations and in Meerwein-Ponndorf-Verley reductions.

ESTERIFICATION

Organic titanates often provide faster reaction rates and higher yields than conventional acid catalysts. A major advantage of using organic titanates is the quality of the excess alcohol recovered from the reactions. Acid-catalyzed methods often cause extensive degradation of the alcohol during the reaction, but titanate catalysis avoids this, allowing recycling of recovered alcohols.

Equally important is the ability of organic titanate catalysis to avoid color formation in the product and to provide high-purity esters suitable for the most demanding applications.

In reactions where color formation remains a persistent problem with titanates, several methods have been reported to minimize it, including the use of phosphorous compounds,^{1,2} mixed metal catalysts,³⁻⁵ organic amine co-catalysts,⁶⁻⁸ and titanium chelates.⁹⁻¹¹

Titanates are superior catalysts compared to most other catalysts because:

- They are not toxic
- They can be removed
- Catalyst residue after removal is not toxic
- High conversion rates, lower cycle time
- High yields on acids and alcohols
- Low byproduct formation
- Low color formation
- Low odor formation

Certain low-quality raw materials may form color with acid-type catalysts. Using organic titanates in the manufacture of trimellitate esters, for example, provides improved color in addition to the higher yield, decreased cycle time and increased product purity.

CATALYST SELECTION

Although Tyzor® TPT contains maximum active titanium per unit weight, it may not always be the most economical choice. Tyzor® TPT is sensitive to water and is highly reactive with alcohols, acids, or other active hydrogen compounds, and may lose isopropyl alcohol quickly by exchange reactions.

In most esterifications, the first reaction is exchange of the isopropyl group with the most active and available materials. Higher alkyl titanates, such as Tyzor® TnBT and TOT, and chelates such as Tyzor® TE, Tyzor® LA and Tyzor® AA-75 are more stable and, in certain cases, may provide a more practical and efficient source of titanium catalyst. Tyzor® LA, because of its unique hydrolytic stability, has been shown to be as reactive as Tyzor® TPT in selected esterification systems.

Titanate catalysts are soluble and may therefore be added to the reaction mixture at any point in the process. Adding the Tyzor® organic titanate catalyst at temperatures above 150 °C (302 °F) may be beneficial, both in efficiency of catalyst usage and in product quality.

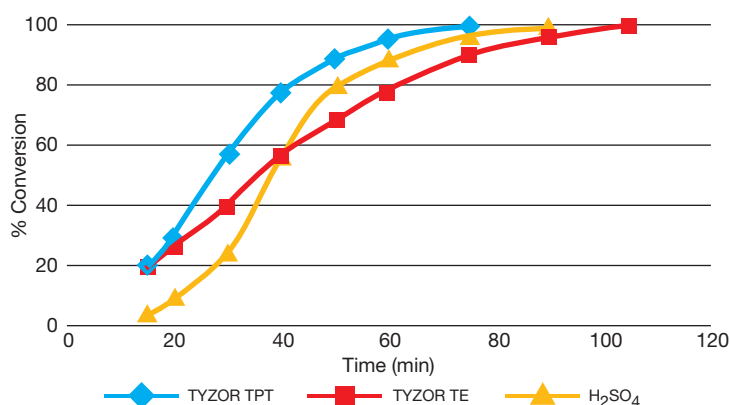
Direct Esterification of Adipic Acid to Bis-2-ethylhexyl Adipate

CATALYST	wt% based on adipic acid	% CONVERSION (min)			
		60	75	90	105
Tyzor® TPT	1.3	96	100	—	—
Tyzor® TE	1.3	78	90	96	100
H ₂ SO ₄	0.6	89	97	100	—

The above example of the preparation of bis-2-ethylhexyl adipate was run with various catalysts as an example Tyzor® catalyst efficacy compared to sulfuric acid-catalyzed reaction esterifications, at about 0.1 wt%, based on the ester formed.

In alcohol-carboxylic acid esterifications, a temperature of 180 °C to 220 °C (356 °F to 428 °F) is usually required to achieve rapid conversion. In esterification it is particularly important that the water resulting from the reaction be removed as rapidly as

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it is formed to avoid deactivating the catalyst by hydrolysis. A polymerization inhibitor and nitrogen blanket are helpful. When an organic titanate catalyst is to be evaluated, all starting materials should be thoroughly dry before the titanate catalyst is added. Drying by azeotropically distilling the water with a suitable solvent is often convenient. About 0.5 wt% of organic titanate, based on the weight of the total product expected, is suggested as a trial catalyst level. In commercial practice, about 0.2 wt% of titanate catalyst is used, based on the weight of the acid (anhydride), in direct monomeric carboxylic esterifications, or about

0.1 wt%, based on the ester formed. In alcohol-carboxylic acid esterifications, a temperature in the range of 180 °C to 220 °C (356 °F to 428 °F) is usually required to achieve rapid conversion. In esterification, it is particularly important that the water resulting from the reaction be removed as rapidly as it is formed to avoid deactivating the catalyst by hydrolysis. A polymerization inhibitor and nitrogen blanket may be desirable if a sensitive ester is being prepared (e.g., an unsaturated ester).

There are many examples of organic titanate usage in industrial esterifications and transesterifications.¹²⁻²¹ Some common reaction systems include phthalic anhydride and 2-ethylhexanol, isophthalic or terephthalic acid and ethylene glycol, and dimethyl terephthalate and ethylene glycol.

REMOVAL OF CATALYST

Titanate catalysts are generally soluble in product esters but are easily removed by conversion to insoluble hydrolysis products, which are then filtered. Titanates may be hydrolyzed using a variety of agents, including aqueous caustic or lime when very low acidity or assistance in color treatment is desired. Generally, agitation with a small amount of water for a period of time before steam stripping is sufficient for efficient hydrolysis of the catalyst. The resulting hydrous titanium oxides are insoluble solids that can be removed by filtration.

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